

Math 330 Abstract Algebra Notes

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1 Group Theory

1.1 Modular Arithmetic

Definition 1.1. Take $n > 0$ and $n \in \mathbb{Z}$. We say that $a, b \in \mathbb{Z}$ are *congruent modulo n* ($a \equiv b \pmod{n}$) if n divides $a - b$ ($n|a - b$) or if the remainder of n/a equals the remainder of n/b .

Here are some properties:

1. if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $a + c \equiv b + d \pmod{n}$.

Proof. $a \equiv b \pmod{n}$ implies that $n|a - b$ which implies that $\exists q \in \mathbb{Z}$ such that $a - b = nq$. The same can be said for $a \equiv b \pmod{n}$ for some $q' \in \mathbb{Z}$ such that $c - d = nq'$. Adding these equations together we have that $a - b + c - d = (a + c) - (b + d) = n(q + q')$. Since the integers are closed under addition, $q + q'$ is an integer and therefore $n|(a + c) - (b + d)$ and as a result, $a + c \equiv b + d \pmod{n}$. $\square$

2. if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $ac \equiv bd \pmod{n}$

Proof. $a \equiv b \pmod{n}$ implies that $n|a - b$ which implies that $\exists q \in \mathbb{Z}$ such that $a - b = nq$. The same can be said for $a \equiv b \pmod{n}$ for some $q' \in \mathbb{Z}$ such that $c - d = nq'$. We can write ac as

$$\begin{aligned} ac &= (b + nq)(d + nq') \\ &= bd + bnq' + dnq + n^2qq' \\ \Rightarrow ac - bd &= n(bq' + dq + nqq') \end{aligned}$$

and since everything in the parentheses are integers and the integers are closed under multiplication and addition, their result is an integer and therefore $n|ac - bd$ and $ac \equiv bd \pmod{n}$. $\square$

3. Congruence mod n is an equivalence relation

Proof. An equivalence relation is reflexive, symmetric, and transitive. To prove reflexivity, we show $a \equiv a \pmod{n} \rightarrow n|a - a \rightarrow n|0$ which is always true. To prove symmetry, we have that $a \equiv b \pmod{n}$ implies $(a - b) = nq$ which we can extract a negative sign such that $(b - a) = n(-q)$ and since $-q$ is also an integer, $n|b - a$ and therefore $b \equiv a \pmod{n}$. To prove transitivity, we say that $a \equiv b \pmod{n}$ implies $a - b = nq$ and $b \equiv c \pmod{n}$ implies $b - c = nr$. We now find that $a - c = b + nq + nr - b = n(q + r)$. Since the integers are closed under addition, $q + r \in \mathbb{Z}$ and therefore $n|a - c$ and $a \equiv c \pmod{n}$. $\square$

1.2 Introduction to Groups

Definition 1.2 (Group). A group is a set of elements G with a binary operation $*$ that assigns to every pair (g, h) of elements of G another element in G denoted $g * h$. Groups follow the group axioms:

1. Associativity

$$(g_1 * g_2) * g_3 = g_1 * (g_2 * g_3)$$

2. There exists an element $e \in G$ that is the identity such that

$$e * g_1 = g_1 * e = g_1$$

3. Every element has an inverse element denoted g^{-1} such that

$$g * g^{-1} = g^{-1} * g = e$$

4. The set of elements is closed under the group operation (although this is implied in the definition).

Example 1.3. Let us look at some examples

1. The set $(\mathbb{Z}, +)$. Addition is associative, the identity is 0, the inverse of each element g is $-g$.
2. The set $(\mathbb{R}_{>0}, \cdot)$ where $\cdot$ denotes regular multiplication. Regular multiplication is associative, the identity is 1, the inverse element is $\frac{1}{g}$.
3. The set $(\{-2, -1, 0, 1, 2\}, +)$. This fails due since it is not closed (take $1 + 2 = 3 \notin$ the set).
4. The set $(\{\frac{1}{2^n}\}, \cdot)$. This fails since there is no inverse.
5. Fix a positive integer n . Take the set $(\{0, 1, 2, \dots, n-1\}, + \text{ mod } n)$. The identity is 0, addition is associative. The inverse of an element $0 \leq a \leq n-1$ is $n-a$ since $(n-a) + a \text{ mod } n = n \text{ mod } n = 0$. This group is denoted by Fraleigh as $\mathbb{Z}_n$ but more generally denoted as $\mathbb{Z}/n\mathbb{Z}$.
6. Fix an $n \in \mathbb{N}$. We denote the group $\text{GL}_n(\mathbb{R})$ as the set of invertible $n \times n$ matrices with real entries with the operation of matrix multiplication. Matrix multiplication is associative, the identity is the $n \times n$ identity matrix I_n . The inverse of every element A is the inverse matrix of A . This group is also known as the *General Linear Group*.

1.3 Properties of Groups

Definition 1.4. The *order* of a group G , denoted $|G|$, is the cardinality of the group/set. If G is finite, then $|G| < \infty$.

Definition 1.5. If the group operation $*$ is commutative, then the group is *Abelian*.

Here are some properties:

1. The identity is unique

Proof. Suppose we have some $f \in G$ such that $f * g = g \ \forall g \in G$. Then if we compose f with the identity, we should get $f * e = e$. By axiom 2 of the group axioms, we must also have that $f * e = f$ and therefore we have that $f * e = e = f$. $\square$

2. Each inverse element is unique.

3. Cancellation, that is if $a * b = c$ for some $a, b, c \in G$, then $b = a^{-1} * c$.

Proof. By group axiom 3, there exists an inverse a^{-1} . We can then left multiply by a^{-1} such that $a^{-1} * a * b = a^{-1} * c$ and by associativity, we then have $b = a^{-1} * c$. $\square$

The consequences of this is that we can make the following observations: $a * b = a * c$ implies that $b = c$ and $b * a = c * a$ implies $b = c$.

For finite groups, we can write a group table where we have the left argument along the rows and the right argument of the operation along the columns. Since Abelian groups are commutative, they have a symmetry across the diagonal. An interesting thing to note with these group tables is that every entry appears only once in each row and column.

Proof. If we have a repeated entry in a row, that implies $a * b = a * c$ but by cancellation, we have that $b = c$. For each column, this implies that $b * a = c * a$ which also implies $b = c$. $\square$

Example 1.6. Let us take a group of cardinality 2. We require that there exists an identity element so the set contains $\{e, a\}$ where a is some arbitrary element. We can immediately fill out the sides adjacent to the header rows by the group axioms (since $e * g = g$) in Table 1. Therefore we must have that $a * a = e$ and therefore a is its own inverse. We shall also note if we apply the group operation n -times to an element, we can represent it as $a^n = a * a * \dots * a$ (n times). If we take a group of cardinality three, we have that $G = \{e, a, b\}$. We once again

	e	a
e	e	a
a	a	e

Table 1: Group table for a group of cardinality 2.

can fill out the header adjacent row and column by composition with the identity. In the $a * b$ cell of Table 2, we require that be e because otherwise, that would require $a * a$ to be e and therefore $b * a = a$ but a was already in that column and so hence we have a contradiction. Therefore we must have $a * a = b$ and $a * b = e$. From this, we can see that $a^2 = b$ and if we multiply $a * b = e$, this implies $a * a^2 = a^3 = e$. Our group is therefore $G = \{e, a, a^2\}$.

	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

Table 2: Group table for a group of cardinality 3

Example 1.7. Let us look at the Quaternion group. The Quaternion group is the set

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$$

under standard multiplication. 1 is the identity element and $i^2 = j^2 = k^2 = -1$. Quaternion multiplication is defined as

$$\begin{aligned} ij &= k & jk &= i & ki &= j \\ ji &= -k & kj &= -i & ik &= -j \end{aligned}$$

This is a group of order 8 and is non-abelian. The inverse of -1 is -1 and so because of this we can say the inverse i and $-i$ and same for j and k . Associativity is also good. This group is the reason why the cross product exists in $\mathbb{R}^3$ and the related Octonion group allows for the existence of the cross product in $\mathbb{R}^7$.

Example 1.8. Another important group is the set of complex numbers with norm of 1 under multiplication. In other words, $\{z \in \mathbb{C} : |z| = 1\}$. This is an Abelian group and is of infinite order. This is a group because the norms are all one so multiplication stays within the group ($|z_1||z_2| = |z_1 z_2| = 1$) and the inverse is also in the group (if $|z_1| = 1$, $\frac{1}{|z_1|} = 1$). We can also convert each complex number into its polar form $z = re^{i\theta}$ $r, \theta \in \mathbb{R}$ and since each element has a norm of 1, $r = 1$ and we can just take $z_1 z_2 = e^{i(\theta_1 + \theta_2)}$ and since the real numbers are a group under addition, this set of complex numbers also form a group. This group is also used to generate groups which contain the n^{th} primitive roots of unity. To do this, we fix a number $n \in \mathbb{N}$. Then the number

$$e^{\frac{2k\pi}{n}i} \quad k \in \{1, 2, \dots, n-1\}$$

is an n^{th} primitive root of unity. This means that these are roots to the equation $z^n = 1$ and the primitive means that n is the smallest power that can satisfy the equation (e.g., no smaller number (let's call m) exists such that $z^m = 1$). We then say that the set of the n^{th} groups of unity form an Abelian group under multiplication of order n . One can see this because multiplication of complex numbers is associative, the identity is 1 ($k = 0$ root)

Definition 1.9. Suppose A is any set. A permutation is a function $f : A \mapsto A$ that is bijective.

Proposition 1.10. *The set of all permutations of a set A is a group where the binary operation $*$ is composition and e is the identity permutation.*

Proof. We must first show that the composition of permutations is also a permutation. Let us say that σ, τ are permutations of A . We can now prove that $\sigma\tau$ is injective. Suppose

$$(\sigma\tau)(a) = (\sigma\tau)(b) \rightarrow \sigma(\tau(a)) = \sigma(\tau(b))$$

Since σ is a permutation, σ is injective and therefore

$$\tau(a) = \tau(b)$$

Using that logic again, τ is a permutation and therefore injective and

$$a = b$$

To show that this is onto, we have to show that for some $a \in A$, there exists an $x \in A$ such that $a = \sigma\tau(x)$. Since σ is onto, there exists a $y \in A$ such that $a = \sigma(y)$. τ is also surjective and so we can say there exists an $x \in A$ such that $y = \tau(x)$. Putting everything together we have that

$$a = \sigma(y) = \sigma(\tau(x)) = (\sigma\tau)(x)$$

As such we have shown that $\sigma\tau$ is a permutation. Now we can check the group axioms. The group operation is associative since function is associative

$$((\sigma\tau)\rho)(a) = \sigma(\tau(\rho(a))) = \sigma(\tau\rho(a)) = (\sigma(\tau\rho))(a)$$

We also denote the identity permutation ι such that $\iota(a) = a \ \forall a \in A$. Finally, any permutation is invertible since it is surjective and given any $y \in A$, there exists only one x such that $y = \sigma(x)$ since σ is bijective. We therefore have that $\sigma^{-1}(y) = x$. We can check that σ^{-1} is a permutation by using the fact that $\sigma(\sigma^{-1}(x)) = x$. Let us consider $\sigma^{-1}(y) = \sigma^{-1}(x)$ and applying σ to both sides we get $x = y$ and hence injectivity. We shall define a $y = \sigma(x)$ and therefore $\sigma^{-1}(y) = \sigma^{-1}(\sigma(x)) = x$ for all $y \in A$. $\square$

This group is known as the symmetric group and we denote it S_A . The most important symmetric group is when $A = \{1, 2, \dots, n\}$. We denote this group as S_n and it has cardinality $|S_n| = n!$.